BDA400 — Data Science Tools and Techniques

Assignment 1: Markdown Buddy

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Submission file: Anne_Agyeman_CA_BDA400_A01.pdf

Project: Synthetic Sales Analysis in R

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Prepared example documentation. GitHub upload, R execution, Markdown preview, and student verification have not been completed.

README.md

Synthetic Sales Analysis in R

Overview

This educational project demonstrates a simple, reproducible workflow for summarizing a **synthetic** sales dataset in R. No real customer or business data is used.

Project Structure

- `sales_analysis.R`: creates synthetic data and calculates summary statistics.

- `sales_analysis.Rmd`: explains the script, its inputs, and its outputs.

- `README.md`: setup and usage documentation.

- `Reflection.md`: reflection on AI-assisted documentation.

- `Appendix.md`: prompts, selected AI responses, and verification checklist.

Requirements

- R (a recent version)

- Base R only; no additional packages required

Installation

1. Install R from https://cran.r-project.org/.

2. Download or clone this repository.

3. Open the project folder in RStudio or another R environment.

Example Usage

Run `source("sales_analysis.R")` in the R console.

```r
source("sales_analysis.R")
```

The script prints a summary table showing total sales by category. All records are generated inside the script.

Inputs and Outputs

Input: a small synthetic data frame containing item category, quantity, and unit price.

Output: a category-level summary with total sales calculated as `quantity * unit_price`.

Verification

Review the source code, run it in R, compare the totals with a manual calculation, and preview this Markdown on GitHub. These checks are **pending user verification**; AI did not execute R or publish the repository.

Limitations

The dataset is illustrative, very small, and unsuitable for drawing real business conclusions.

License

Educational demonstration; no license has been selected. Add a license file before distributing the project under specific reuse terms.

AI Assistance Disclosure

Tool: ChatGPT (GPT-6), 2026-10-22. Prompts requested a README outline, concise Markdown formatting, a synthetic-data project README, an R Markdown structure, and a syntax review. AI helped draft and organize text. The student must inspect the code, run the script, check calculations, preview Markdown, and record any edits. See `Appendix.md` for prompts and sample responses.

AI Assistance Declaration

AI Assistance Declaration: I used ChatGPT (GPT-6) for documentation structure, draft wording, and sample code layout. Prompts used: see Appendix.md. Verification methods to complete: R execution, manual arithmetic cross-check, Markdown preview, and self-review. All final calculations must be verified by me. I am responsible for the accuracy and originality of this work.

sales_analysis.R

```
# Synthetic sales example (base R only)
sales <- data.frame(
  category = c("Books", "Books", "Games", "Games"),
  quantity = c(2, 3, 1, 4),
  unit_price = c(10, 10, 20, 20)
)
sales$total <- sales$quantity * sales$unit_price
summary <- aggregate(total ~ category, data = sales, FUN = sum)
print(summary)
```

sales_analysis.Rmd

```
---
```

title: "Documenting a Synthetic Sales Analysis Script"

author: "Anne Agyeman"

output: html_document

```
---
```

AI Assistance Declaration

I used ChatGPT (GPT-6) on 2026-10-22 for a documentation draft and code-block formatting. See `Appendix.md` for prompts. R execution, manual verification, and knitting are pending. I am responsible for checking all final work.

Purpose

This document explains how `sales_analysis.R` generates and summarizes synthetic sales records.

Inputs

The script uses a synthetic, in-memory table with `category`, `quantity`, and `unit_price`. No external file is needed.

Method

The code computes row-level sales and aggregates totals by category.

```
```{r sales-analysis}
sales <- data.frame(
  category = c("Books", "Books", "Games", "Games"),
  quantity = c(2, 3, 1, 4),
  unit_price = c(10, 10, 20, 20)
)
sales$total <- sales$quantity * sales$unit_price
summary <- aggregate(total ~ category, data = sales, FUN = sum)
print(summary)
```
```

Outputs

The output is a two-row category summary printed in the knitted document. Check its numbers by hand before submission.

Validation and Limitations

Open this file in RStudio/Posit Cloud and knit it. Check code chunk execution and rendered headings. These actions have not been performed by the AI. The synthetic sample does not represent actual sales performance.

Reflection.md

Reflection on AI-Assisted Documentation

AI Assistance Declaration

I used ChatGPT (GPT-6), 2026-10-22, to draft reflection examples. Prompts and verification status appear in Appendix.md. I must revise these responses to reflect my actual experience.

Note: The supplied assignment mentions three reflection questions but does not print their wording. The following are three suggested reflection topics, not quotations from the instructor.

1. How did AI help the documentation process?

AI provided a starting structure for the README and R Markdown document. It suggested clear headings for project purpose, requirements, inputs, outputs, and usage. This helped organize information that a new reader would need.

2. What are the limitations of AI-generated documentation?

AI may invent project details, dependencies, outputs, or verification claims. Its formatting may look correct while the R code fails. I need to compare the draft with my actual project and test each claim instead of accepting it automatically.

3. How can AI be used ethically and responsibly?

I should disclose the tool, version, date, and prompts; avoid private data; use synthetic examples; verify outputs myself; and make substantive edits. I should not claim that I ran tests, knitted a file, or uploaded to GitHub unless I did so.

Appendix.md

Appendix: AI Prompts, Responses, and Verification

AI Assistance Declaration

ChatGPT (GPT-6), 2026-10-22, helped draft these materials. All execution and visual verification remain the student's responsibility.

Prompts and Key Responses

1. **Seed:** "Explain what sections a good GitHub README for an R data analysis project should include." **Response summary:** Overview, requirements, installation, project structure, example usage, inputs/outputs, verification, limitations, license, and AI disclosure.

2. **Refinement:** "Revise the sections list so it’s concise and uses Markdown headers and bullet formatting." **Response summary:** Organized the sections with `#`/`##` headings and bullets.

3. **Critique:** "Check the Markdown syntax for correctness and readability." **Response summary:** Use fenced code blocks, consistent headings, and explicit validation notes.

4. **README draft:** "Here’s a summary of my R project: a base-R analysis of four synthetic sales records. Generate a professional README.md file using Markdown." **Response summary:** Drafted a structured README with setup, inputs, outputs, and usage.

5. **README refinement:** "Add sections for Installation, Example Code, and License. Keep tone concise and professional." **Response summary:** Added those sections and noted that a license has not been selected.

6. **README critique:** "Review the Markdown for syntax errors and suggest 2 improvements for clarity." **Response summary:** Clearly state synthetic-data scope and distinguish proposed checks from completed checks.

7. **Rmd seed:** "Here’s my R script description: create four synthetic sales records and aggregate revenue by category. Suggest Markdown formatting and code block examples." **Response summary:** Suggested purpose, inputs, method, outputs, and an R code chunk.

8. **Rmd refinement:** "Add syntax highlighting and improve section organization (e.g., # Purpose, ## Inputs, ## Outputs)." **Response summary:** Used R Markdown chunk syntax and consistent headings.

9. **Rmd critique:** "Is the Markdown consistent with RMarkdown best practices?" **Response summary:** Include YAML, valid fenced chunks, clear headings, and knit verification.

Student Verification Checklist (not yet completed)

- [ ] Read the GitHub Markdown guide: https://github.github.com/gfm/

- [ ] Compare README structure with a real tidyverse repository README.

- [ ] Open README.md in GitHub preview and inspect headings, bullets, and code fences.

- [ ] Run sales_analysis.R in R or RStudio.

- [ ] Independently calculate each synthetic record's sales and category totals.

- [ ] Knit sales_analysis.Rmd in RStudio or Posit Cloud.

- [ ] Review and personalize the three reflections against the actual instructor questions.

- [ ] Create GitHub repository `markdown-buddy-anne-agyeman` and upload project files.

- [ ] Add screenshots or notes of successful verification if requested.

Two Suggested Refinements

1. Add the exact R version tested after running the script.

2. Add a screenshot or copied output from the successful R Markdown knit after verifying it.